

Assignment ÷ 05

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Q1. Solution ÷ There are $4!$ ways to place four different coloured tiles in a row, assuming that the order of the tiles matters.

Any one of the four tiles can be placed in the first space; any one of the three remaining tiles can be placed in the second space; any one of the two remaining can be placed in the third space and the remaining tile can be placed in the fourth space.

By multiplication principle, the number of arrangement is $4 \times 3 \times 2 \times 1 \Rightarrow 24$ Ans

Q2. Solution ÷ Using the multiplication principle, Any one of five tiles can be placed in the first space, any one of the 4 remaining tiles in the second space and any one of three remaining tiles in the third space. So, there are $5 \times 4 \times 3 = 60$ different arrangement or permutation of 5 coloured tiles taken 3 at a time. The number of permutation of 3 objects from set of 5 objects is abbreviated as 5P_3 and we have shown that

$${}^5P_3 = 60 \quad \underline{\text{Ans}}$$

Q3. Solution: using the permutation formula for 7 items taken 3 at a time.

$${}_7P_3 = \frac{7!}{(7-3)!} = \frac{7!}{4!} \Rightarrow \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1}$$

$$\Rightarrow 7 \times 6 \times 5 \\ \Rightarrow 210 \quad \checkmark$$

Q4. Solution: (i) There is no requirement to consider the order of the people on the 3-person Hules committee.

So the number of different committees can be found with the formula for combinations.

$${}_{10}C_3 = \frac{10!}{(10-3)! 3!} \Rightarrow \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1)}$$

$$\Rightarrow \frac{10 \times 9 \times 8}{6} \Rightarrow 120 \quad \checkmark$$

(ii) Order must be considered in choosing the three officers because it makes a difference as to, whom holds each office. So the number of different possibilities for the three office can be found with the formula for permutations.

$${}_{10}P_3 \Rightarrow \frac{10!}{(10-3)!} \Rightarrow \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$\Rightarrow 10 \times 9 \times 8 \Rightarrow 720 \quad \checkmark$$

Q5. Solution: Let the number of elements be N. Thus, the number of permutations is $N!$. If we reduce the number of elements by two, the number of elements becomes $N-2$. The number of permutations, equal to $(N-2)!$, reduces thirty times.

$$\text{Hence, } \frac{N!}{(N-2)!} \Rightarrow 30$$

$$\Rightarrow {}_N C (N-1) = 30$$

$$N = 6 \quad \checkmark$$

Ques 6. Solution: If you have 25 letters to choose from and you're forming a registration number with a pattern of two letters, three numbers and two letters, you have:

$$\Rightarrow 25^4 \times 10^3 \quad \checkmark$$

Q. 7. Solution: Permutation through three square:

Using the combination formula $\left(\frac{n}{k}\right) \Rightarrow \frac{n!}{k!(n-k)!}$

$$\left(\frac{9}{3}\right) \Rightarrow \frac{9!}{3!(9-3)!} \Rightarrow \frac{9 \times 8 \times 7}{3 \times 2 \times 1}$$

$$\Rightarrow 84$$

Permutation through four squares:

$$\left(\frac{9}{4}\right) \Rightarrow \frac{9!}{4!(9-4)!} \Rightarrow \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1}$$

$$= 126$$

Now, adding

$$84 + 126 \Rightarrow 210$$

Q.8. Solution: Using Combination formula.
The number of ways to choose 9 couples from 20 students is given by -

$$\left(\frac{20}{2}\right) \Rightarrow \frac{20!}{2!(20-2)!} = \frac{20 \times 19}{2 \times 1}$$

$$\Rightarrow 190$$

Q.9 Solution: The number of player (N)

$$N \Rightarrow \frac{1 + \sqrt{1 + 8 \times \text{No. of matches}}}{2}$$

for 21 matches:

$$N = \frac{1 + \sqrt{1 + 8 \times 21}}{2}$$

$$N = \frac{1 + \sqrt{169}}{2} \Rightarrow \frac{1 + 13}{2} = \frac{14}{2}$$

$$N \Rightarrow 7$$

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Q.10 Solution: If each of the six married couples wants to sit next to each other, you can treat each couple as a single unit. This gives 6 unit to arrange.

The number of ways to arrange these 6 unit is 6!

$$6! \Rightarrow 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$\Rightarrow 64 \times 720$$

$$\Rightarrow 46,080$$

Q.11 Solution: Note that the Martian may select 4 socks without a pair: one red, one blue, one yellow and one black socks. However, if Martian select 5 socks from 4 colours, the pigeonhole principle guarantees that 2 socks must have the same color.

Therefore, 5 is the minimum number of socks they must draw to guarantee a pair.

Q.12 Solution: Note that for any such a and b, $n \mid (a-b)$

We may rewrite this in modular arithmetic as $a-b \equiv 0$ or $a \equiv b$.

By pigeonhole principle, there exist two integers with the same remainder modulo n. As shown, this implies that the difference is a multiple of n.

Q13

Solution - By pigeonhole principle, let's consider the seven line segments.

If you have seven line segments, each with length no greater than 10 inches and no shorter than 1 inch, you can group them based on their lengths.

So, By the pigeonhole principle, if you have 7 line segments, at least 3 of them will have the same length, and you can choose those three to represent the side of a triangle.

Q14

Solution - In the worst case, consider that senator of 5 hates a set of 3 senators, while he himself is hated by a completely different set of 3 other senators. Thus, given one senator, there may be a maximum of 6 other senators whom he can't work with. If we have a minimum of 7 committees, there should be at least one committee suitable for the senator 5 after the assignment of the 6 conflicting senators.

Q15

Solution - We have 12 months, so $100/12$
 $\Rightarrow 8.33$ gives us the content of each month. But we can't have partial people in each month, so we round up:

$$\left\lceil \frac{100}{12} \right\rceil = 9$$

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Solution - This follows because when 21 objects are distributed into 10 boxes, one must have $\left\lceil \frac{21}{10} \right\rceil = 3$ elements.

We have $K=5$ and $M=6$, we need to compute N such that $\left\lceil \frac{N}{K} \right\rceil = M$ or more precisely

$$\left\lceil \frac{N}{K} \right\rceil = 6.$$

We can compute the smallest integer with this property as $N = K(M-1) + 1$. Plugging K and M into this equation gives us $N = 5(6-1) + 1 = 26$.

Thus, 26 is the minimum number of students needed to ensure that at least six students will receive the same grade.

Q18

Solution - There are eight million different phone numbers of the form $NYX-XXXX$. Hence, by the generalized pigeonhole principle, among 25 million telephones, at least

$$\left\lceil \frac{25000000}{8000000} \right\rceil = 4 \text{ have of them}$$

must have identical phone numbers. Hence, at least four area codes are required to ensure that all 10 digit numbers are different.

Q19. Solution :

(a). $a_n = 3a_{n-1}$

$\Rightarrow$ It is linear homogeneous since coefficient with a degree of 1.

(b). $a_n = 2na_{n-1}$

$\Rightarrow$ It is not linear homogeneous since it has a term with n in front.

(c). $a_n = 6a_{n-1} + 7$

$\Rightarrow$ It is linear homogeneous of constant coefficient with a degree of 1.

(d). $a_n = 3a_{n-1} - 2a_{n-2}$

$\Rightarrow$ It is linear homogeneous of constant coefficient with a degree of 2.

(e). $a_n = a_{n-4}a_{n-2}$

$\Rightarrow$ It is not a linear homogeneous due to the product of term involving different indices.

(f). $a_n = 3na_{n-1}$

$\Rightarrow$ It is not linear homogeneous as it has a term with n in front.

(g). $a_n = a_{n-1}^2 + a_{n-2}$

$\Rightarrow$ It is not linear homogeneous due to the square term.

Q20. Solution : Let's verify that $a_n = 0$, $a_n = 4^n$, $a_n = n \cdot 4^n$, and $a_n = 2 \cdot 4^n + 3n \cdot 4^n$ are the solⁿ of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$.

1. For $a_n = 0$;

$$0 = 8 \cdot 0 - 16 \cdot 0$$

This is true, so $a_n = 0$ is a solution.

2. For $a_n = 4^n$;

$$4^n = 8 \cdot 4^{(n-1)} - 16 \cdot 4^{(n-2)}$$

Simplifying both sides confirms that this holds true. So $a_n = 4^n$ is a solution.

3. For $a_n = n \cdot 4^n$;

$$n \cdot 4^n = 8 \cdot (n-1) \cdot 4^{(n-1)} - 16 \cdot (n-2) \cdot 4^{(n-2)}$$

This is true.

4.

For $a_n = 2 \cdot 4^n + 3n \cdot 4^n$;

$$2 \cdot 4^n + 3n \cdot 4^n = 8 \cdot (2 \cdot 4^{(n-1)} + 3(n-1) \cdot 4^{(n-1)}) - 16 \cdot (2 \cdot 4^{(n-2)} + 3(n-2) \cdot 4^{(n-2)})$$

After simplifying, it holds true.

Therefore, all four expressions are solutions to the recurrence relation.

Ques 21 :- Solution : Let's substitute $a_n = c_1 + c_2 \cdot 2^n - n$ into recurrence relation $a_n = 3a_{n-1} + 2a_{n-2} - 1$

$$\Rightarrow (c_1 + c_2 \cdot 2^n - n) - 3(c_1 + c_2 \cdot 2^{n-1}) - (n-1) + 2(c_1 + c_2 \cdot 2^{n-2} - (n-2))$$

$$\Rightarrow c_1 + c_2 \cdot 2^n - n - 3c_1 - 3c_2 \cdot 2^{n-1} + 3(n-1) + 2c_1 + 2c_2 \cdot 2^{n-2} - 2(n-2)$$

$$\Rightarrow -2c_1 - c_2 \cdot 2^n + 3 + 2c_1 + c_2 \cdot 2^n - 1$$

After simplifying we get 1, confirming that $a_n = c_1 + c_2 \cdot 2^n - n$ is indeed a solution of the given recurrence relation $a_n = 3a_{n-1} + 2a_{n-2} - 1$.

Q.22. Solution : Let's substitute an into the left side of the recurrence relation.

$$\Rightarrow c_1 \cdot 2^n + (c_2 \cdot 4^n) - 6(c_1 \cdot 2^{n-1} + c_2 \cdot 4^{n-1}) + 8(c_1 \cdot 2^{n-2} + c_2 \cdot 4^{n-2})$$

$$= c_1 \cdot 2^n + c_2 \cdot 4^n - 6c_1 \cdot 2^{n-1} - 6c_2 \cdot 4^{n-1} + 8c_1 \cdot 2^{n-2} + 8c_2 \cdot 4^{n-2}$$

$$= 2^n (c_1 - 6c_1 + 8c_1) + 4^n (c_2 - 6c_2 + 8c_2)$$

$$= 0$$

After simplifying, we get 0. Hence proved.

Q.23 Solution : Let's find the first six terms of the sequence $a_n = \frac{n-1}{n+1}$ by substituting value for n:

$$a_1 = \frac{1-1}{1+1} = 0, \quad a_2 = \frac{2-1}{2+1} = \frac{1}{3}$$

$$a_3 = \frac{3-1}{3+1} = \frac{1}{2}, \quad a_4 = \frac{4-1}{4+1} = \frac{3}{5}$$

$$a_5 = \frac{5-1}{5+1} = \frac{2}{3}, \quad a_6 = \frac{6-1}{6+1} = \frac{5}{7}$$

Now, let's show that $a_{n+1} - a_n = \frac{2}{(n+1)(n+2)}$

$$a_{n+1} - a_n = \frac{(n+1)-1}{(n+1)+1} - \frac{n-1}{n+1}$$

After simplifying

$$\left\{ a_{n+1} - a_n = \frac{2}{(n+1)(n+2)} \right\} \text{ Proved.}$$

Q.24.

Solution

$$a_n = a_{n-1} + 2; \quad a_0 = 1$$

$$a_1 = a_0 + 2 = 1 + 2 = 3$$

$$a_2 = a_1 + 2 = 3 + 2 = 5$$

$$a_3 = a_2 + 2 = 5 + 2 = 7$$

And so on. The solution is $[a_n = 2n+1]$

b) $a_n = 2a_{n-1} + 1$; $a_1 = 7$
 $a_2 = 2 \cdot 7 + 1 \Rightarrow 15$
 $a_3 = 2 \cdot 15 + 1 \Rightarrow 31$
 $a_4 = 2 \cdot 31 + 1 \Rightarrow 63$
 $[a_n \Rightarrow 2^{n+2} - 1]$

c) $a_n = a_{n-1} + n$; $a_1 = 4$
 $a_2 = 4 + 2 = 6$
 $a_3 = 6 + 3 = 9$
 $[a_n = \frac{n(n+1)}{2} + 3]$

d) $a_n = n a_{n-1}$; $a_0 = 5$
 $a_1 = 5 \cdot 1 = 5$
 $a_3 = 2 \cdot 5 = 10$
 $a_4 = 3 \cdot 10 = 30$
 $[a_n = 5n!]$

Q.25 Solution: Let's substitute $a_n = -2^{n+1}$ into the recurrence relation:

$a_n = 3a_{n-1} + 2^n$
 $-2^{n+1} = 3(-2^{(n+1)+1}) + 2^n$
 After simplifying,

$-2^{n+1} = -3 \cdot 2^n + 2^n$

Combining the terms on the right side:

$-2^{n+1} = -2 \cdot 2^n$

It can be observed that the left side is indeed equal to the right side.

Therefore, $a_n = -2^{n+1}$ is a solⁿ of the given non-homogeneous recurrence relation $a_n = 3a_{n-1} + 2^n$

Q.26 Solution: Let's substitute a_n into recurrence relation:
 $a_{n+2} = 5a_{n+1} + 6a_n = \frac{1}{6} \cdot 5^{n+2} - 5 \cdot \frac{1}{6} \cdot 5^{n+1} + 6 \cdot \frac{1}{6} \cdot 5^n$

Now, let's simplify each term:

Factor out common terms:

$\frac{1}{6} \cdot 5^{n+1} (5 - 5 + 1)$

simplify further:

$\frac{1}{6} \cdot 5^{n+1} \cdot 1$

Therefore, $a_n = \frac{1}{6} \cdot 5^n$ is a solution to the recurrence relation. $a_{n+2} = 5a_{n+1} + 6a_n = 2^n$.

Q.27 Solution: The generating function for the sequence $1, a, a^2, a^3, \dots$ is given by:

$G(x) = 1 + ax + a^2x^2 + a^3x^3 + \dots = \sum_{n=0}^{\infty} a^n x^n$

This is a geometric series with the common ratio a , and the sum can be expressed as a closed form:

$$G(x) = \frac{1}{1-a}$$

So, the generating function for the given sequence is

$$\frac{1}{1-a}$$

Q28. Solution :-

a) $a_n = 3a_{n-1} + 2$; $a_0 = 1$

$\Rightarrow G(x) = a_0 + a_1 x + a_2 x^2 + \dots$

Multiplying both sides by x^n and summing over all n

$$G(x) = a_0 + \sum_{n=1}^{\infty} (3a_{n-1} + 2) x^n$$

Rearranging terms and using the property $a_n = 3a_{n-1} + 2$ we get:

$$G(x) = 1 + 3x G(x) + 2 \sum_{n=1}^{\infty} x^n$$

$$G(x) = \frac{1}{1-3x}$$

b)

$$a_n - 9a_{n-1} + 20a_{n-2} = 0 ; a_0 = -3, a_1 = -10$$

$\Rightarrow$

$$G(x) = a_0 + a_1 x + a_2 x^2 + \dots$$

$$G(x) = a_0 x^0 + a_1 x^1 + \sum_{n=2}^{\infty} (a_n - 9a_{n-1} + 20a_{n-2}) x^n$$

Rearranging terms:

$$G(x) = -3 - 10x + \sum_{n=2}^{\infty} (a_n - 9a_{n-1} + 20a_{n-2}) x^n$$

$$G(x) = -3 - 10x + \sum_{n=2}^{\infty} 0 x^n$$

$$G(x) = \frac{-3 - 10x}{1 - 9x + 20x^2}$$

c)

$$a_{n+2} - 2a_{n+1} + a_n = 12 ; a_0 = 2, a_1 = 1$$

$\Rightarrow$

$$G(x) = a_0 + a_1 x + a_2 x^2 + \dots$$

$$G(x) = a_0 x^0 + a_1 x^1 + \sum_{n=2}^{\infty} (a_{n+2} - 2a_{n+1} + a_n) x^n$$

$$G(x) = 2 + x + \sum_{n=2}^{\infty} 12 x^n$$

$$G(x) = \frac{2+x}{1-2x+x^2}$$

* Completed *

THANK YOU !